

A Review of UAV Ferry Algorithms in Delay Tolerant Network

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Abstract—The Delay Tolerant Network (DTN) is suitable for post-disaster environments where infrastructure is scarce. Search and rescue operations require network deployment in areas not covered by infrastructure. In the post-disaster DTN environment, UAVs are introduced as ferry nodes for message-passing and it is a popular practice recently. This paper introduces the application of Delay Tolerant Network in post-disaster environments and introduces message-passing through ferry nodes in DTN. Next, this paper analyzes the UAV's ferry algorithms from the perspectives of UAV network architecture, node routing algorithm, node deployment, node trajectory optimization, node energy optimization and node storage allocation optimization in DTN. The application of UAV ferry algorithm in post-disaster DTN environment is analyzed, and the performance evaluation as well as the comparison of different algorithms are given. Finally, this paper looks forward to the future research direction of UAV ferry algorithm in DTN environment.

Keywords—Delay Tolerant Network; UAV; ferry node; algorithm

I. INTRODUCTION

In the disaster area environment, battlefield environment and interstellar communication environment, infrastructure is scarce or even damaged due to bad disasters, resulting in sparse communication nodes and limited communication power. Search and rescue operations require network deployment in areas not covered by infrastructure. Traditional mobile ad hoc network is based on the TCP/IP protocol [1]. Under such limited network conditions, high real-time performance cannot be ensured, and the use of traditional TCP/IP-based protocols may result in network performance degradation or even failure to work properly.

Delay Tolerant Network (DTN) is suitable for this limited network environment and it is a sparse network where the complete end-to-end path between the source and the destination does not always exist as expected. At the same time, the path of the nodes is usually difficult to establish, and the data transmission in the network always has a large delay. Data transmission may occur only when the current node is within the communication range of the next hop node. Therefore, when there is data arriving at the node or the node has data to send, the data will first be stored in the cache of the node, and will be carried until the appropriate next hop node is encountered. The data transmission mode is called the "storage-carry-forward" mode. In order to achieve this special transmission mode and maintain certain compatibility with the traditional network,

DTN adds a Bundle layer protocol stack to the traditional Internet protocol stack, which is located above the transport layer and below the application layer to meet the needs of data transfer and caching [2]. By adding a Bundle layer, the communication node effectively overcomes the bad effects of frequent interruptions and high latency of network connections. The routing mechanism of the DTN relies on the mobility of the nodes to connect to the disconnected nodes, passing messages around the network to overcome path disconnection. The development of DTN provides a more reliable communication guarantee for many future military wars, space communications, disaster relief and other scenarios.

II. OVERVIEW OF UAV FERRY NODES IN DTN

Nowadays, the introduction of ferry nodes in DTN for message transmission between different regions is a hot research issue. Unmanned Aerial Vehicles (UAV) has been a research hotspot in DTN environment due to its low cost, rapid together with flexible deployment and strong mobility. This research direction is widely followed by researchers.

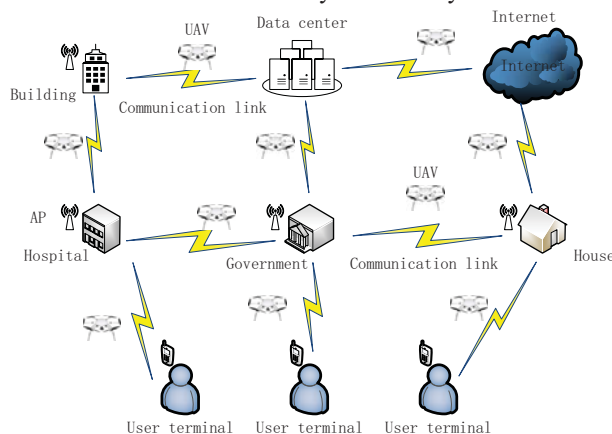


Figure 1. UAV ferry nodes in DTN.

UAVs can provide timely warnings of disasters as well as assist in accelerating search and recovery work when the large public communication network is down. Large UAVs can perform missions alone, while small UAVs are suitable for formation flights. Compared to a single UAV system, a UAV system consisting of multiple UAVs can collaborate more efficiently and economically. However, the routing requirements of the UAV network exceed the routing requirements of traditional mobile ad hoc networks. They need to adapt to protocols with strong mobility, dynamic

topology changes, intermittent link connections, and tight power constraints. At the same time, UAVs may be powered off or even going wrong, and the network may be divided into several partitions, making how to handle delays and interrupts a necessary design consideration. The protocol design of each layer should also help improve the web environment. UAV can operate as a data ferry node in a post-disaster DTN, acting as a mobile relay between multiple partition nodes, and performing data transmission. The application of the UAV ferry nodes in the post-disaster DTN is shown in Figure 1.

III. UAV FERRY NODE ALGORITHMS IN DTN

In a UAV network, the UAV communication routing protocol is not a simple implementation of an active or passive solution. When some UAVs fail, the backbone network between the UAVs must be reorganized repeatedly [3]. UAVs move in 3D scenes, which caused their locations change rapidly and the network topology changes frequently. These factors may lead to new connections or disconnections. Therefore, it is significant to design an effective algorithm for the UAV ferry nodes in DTN to cope with these new challenges.

A. UAV network architecture in DTN

The performance of the UAV network changes with the surrounding environment, and a good UAV network architecture is critical to improving network performance. By combining the advantages of MANET and DTN, Huang et al. [4] proposed an adaptive network architecture to improve data transmission performance of UAV network in unreliable networks. All source nodes select the most appropriate architecture in a centralized manner based on the state of the surrounding links and the size of the packets that need to be sent. The intermediate node does not need to make any changes during the transfer. This architecture significantly improves the data transfer performance of the UAV network.

Setianingsih et al. [5] proposed an ION-DTN emergency communication method based on UAV system, using an UAV system to construct an ION-DTN network for data transmission. The UAV system is equipped with cameras together with artificial intelligence devices to identify the affected people, and the images or locations of the affected people are transmitted from different source nodes to server through ION-DTN. The optimal delivery rate of the scheme in the experimental environment is hundred percent, the minimum delay time is 37.7s, the minimum jitter time is 0.534s, and the maximum throughput is 69.5kbps, which effectively improves the performance of the network.

B. UAV ferry algorithms

In the UAV ferry algorithms in the DTN environment, the UAV ferry routing algorithm plays an important role in how to efficiently transfer data from source to destination through the ferry nodes. Literature [6-8] introduce auxiliary nodes for routing decisions. Xiong et al. [6] proposed a routing policy based on regional central nodes (CPR), and placed throw boxes as fixed regional central nodes in the center of the personnel gathering area. This realizes data

transmission between nodes in the area and the ferry nodes. Barroca et al. [7] introduced a cut-off trigger pigeon with a deadline traveler problem (DTP-TSP-D), and the secondary node acts as a shared messenger by collecting and transmitting messages on the cluster of nodes. Once the node is triggered, it becomes a pigeon node. The pigeon node is split to pass messages outside of the specified cluster. Shin et al. [8] proposed a progressive relay node placement route reconstruction algorithm. Among them, global UAVs are used to traverse and detect advanced information in longer steps, while using local UAVs to traverse and detect local information in shorter steps. During the best effort full distributed traversal, the global UAVs and local UAVs encountered can collaborate to find more optimized deployment locations for routing.

Literature [9-11] make routing decisions based on the location information of nodes. Arafat et al. [9] proposed a LADTR protocol. The future position of the UAV is estimated based on its current position and its speed. The forward UAV predicts the location of the destination UAV and then decides where to forward. Bujari et al. [10] proposed an airborne DTN search and rescue scene routing protocol (GeoSaW) based on location-aware waypoints, using the node's current location information and mission planning path. The GeoSaW protocol predicts the future location of each relay node and the time at which it arrives. The node waits for the recipients and sends the packet to these recipients, which will carry the message to the destination. Chen et al. [11] proposed a location-based ferry nodes DTN routing algorithm (FRLI). By designing a progressive data transmission strategy based on location information, the ferry nodes are effectively allocated in the network and exchanged with each other. The cached network node information is used to establish a node area attribute table. When the destination node of the data does not belong to the current area, the data is directly delivered to the gateway node, thereby improving data transmission efficiency and reducing the average data transmission delay.

The literature [12-15] introduce other routing decision-making methods. Giannini et al. [12] considered using UAVs to transmit data generated in isolated buildings to regional concentrators, for the purpose of finding the best flight time to maximize delivery rates. Agussalim et al. [13] improved a message transmission scheduling scheme called TTL-Loc and tested it in both 1s-to-2d and 2s-to-1d schemes. In the case of 1s-to-2d, the improved TTL-Loc scheme has better performance and more messages are delivered. As the message load increases, this advantage also increases; in the case of 2s-to-1d, the improved TTL-Loc scheme can achieve better performance only under high load. Based on DTN Routing Hierarchical Topology (DRHT), Alaoui et al. [14] proposed a model that allows for efficient access planning for clusters to avoid overloading the ferry route that causes network failure. He et al. [15] proposed MRA algorithm that classifies nodes based on the Mean Weighted Moving (MWM) to deliver a copy of the message to a node with a higher MMW. When performing single-copy transmission, the selection of the relay node needs to consider historical contact information with the destination node in addition to

the MWM. The algorithm effectively reduces the network load and improves the message delivery rate.

The literature [16-18] introduce the deployment algorithms of the UAV ferry nodes. Kalantari et al. [16] proposed a heuristic algorithm based on particle swarm optimization, which finds the minimum deployment number of UAVs and their deployment location that can give service to specific areas with different user densities. However, the author did not jointly consider the coverage of the UAVs and the connectivity between the UAVs. Wang et al. [17] comprehensively considered the connectivity and stability of the UAV network, and proposed a new minimum UAV deployment algorithm, which models the UAV deployment as a target optimization function. The scheme initially selects enough candidate locations to assume that the UAVs are deployed there, then iteratively removes the redundant locations by heuristic algorithm, and checks the UAV link of the remaining locations during the deletion process. The last remaining is the necessary locations, and then deploy the UAVs in the necessary locations. Han et al. [18] first proposed research on the deployment of UAV relay systems, using the proposed new UAV deployment algorithm to maximize the average data rate with a minimum number of UAVs and a minimum transmission time slot.

The literature [19-21] introduce the trajectory optimization algorithms for the UAV ferry nodes. Harounabadi et al. [19] used genetic algorithm as a heuristic search and approximation of the best solution to plan the path of multiple UAVs. They proposed a network model, which allows multiple transmissions of messages through the aggregation node, thereby achieving message exchange between any two nodes in the network and optimizing the average weighted delay of message delivery. Sambo et al. [20] designed the energy minimization UAV trajectory algorithm based on genetic algorithm, which minimizes the energy demand of the UAV, and its total energy demand is less than two-hop relay multi-session routing algorithm [21] and nearest neighbor trajectory algorithm.

In the literature [22-25], the energy optimization algorithms of the UAV ferry nodes are introduced for the problem of large energy consumption in the UAV ferry nodes. Dong et al. [22] proposed an information collection and forwarding system based on fixed-wing UAVs. The UAV communicates with the two ground nodes in turn. It uses the runway trajectory model to collect and forward data between the ground nodes. The path discretization method and the continuous convex algorithm are jointly used to optimize the trajectory length or radius to minimize energy consumption. Yanai et al. [23] discussed the UAV flight plan for the battery exhaustion in DTN to reduce delay time. The author proposes two options: in the rally flight scenario, UAVs entering the communication range change their destinations to the relative positions; in the case of a common flight path, the UAVs fly on a common flight path to prevent the network path from being disconnected. Niitsu et al. [24] proposed the use of a fixed postal node. Only the postal node can communicate with the ferry node, and they propose two types of messaging. The one is from the mobile

node to the postal node, the other is from the postal node to the mobile node. Among them, the static method and the dynamic method for determining the valid time of the trigger signal effectively reduce the energy consumption of nodes and prolong the available communication time. Nakamura et al. [25] put forward a MultistageStdDev disaster time communication system and introduced an evaluation method that effectively utilizes node energy for communication by setting a multi-level threshold of residual power.

Literature [26] shows that when there are multiple data nodes in the communication range of the DTN ferry node, some data needs to be sent, but the memory of the current ferry node is not enough to save all the data transmission requirements. Therefore, the storage allocation optimization algorithm of the UAV ferry node is introduced. Cong et al. [26] used the weighted minimum and maximum fairness principle to allocate storage space. The ferry node in DTN arrives at the data collection point to obtain a data forwarding request packet in the collection area. If the data storage requirements are less than or equal to the storage resource of the ferry node, the normal data storage and forwarding service is provided for all nodes. Otherwise, it is necessary to optimize the memory space allocation. The performance of the storage resource optimization ferry algorithm has significantly improved the performance in terms of network transmission success rate and average network delay.

IV. COMPARATIVE ANALYSIS

The performance evaluation index in DTN is of great importance for measuring the performance of related algorithms. Since the key of the algorithm is to forward the messages from the source to the destination, the delivery rate of the message is the primary evaluation index of the performance of the algorithm. At the same time, the network transmission delay of the data packet is also an important performance evaluation indicator. Therefore, how to balance the message delivery rate, the network transmission delay and the network resource consumption caused by message forwarding is the primary problem that the UAV ferry algorithm needs to solve. Next, this paper mainly compares the UAV ferry algorithms in DTN from the factors of transmission rate, packet transmission rate, message delivery rate, transmission delay, cost ratio and application scenario. The existing related work summary is shown in TABLE I.

As can be seen from the TABLE I, the focus of the above schemes is different. The balance between the message delivery rate, transmission delay and cost ratio in DTN is a research point that needs to be focused on. Different UAV ferry algorithms have their own advantages and disadvantages. They have been optimized and improved the indicators of their respective concerns, and they all have their certain practical application value. However, because the focus of different programs is different and they have different scenarios for optimization, it is difficult to give a quantitative comparative analysis. Therefore, this paper only conducts qualitative analysis and comparison.

TABLE I. COMPARISON OF UAV FERRY ALGORITHMS IN DTN

Name	Transmission rate	Packet transmission rate	Message delivery rate	Transmission delay	Cost ratio	Application scenario
ION-DTN[5]	—	—	High	Low	—	Post-disaster DTN environment
CPR[6]	—	—	High	Low	Medium	Communication between disaster area
DTP-TSP-D[7]	High	—	—	Medium	—	DTN environment
LADTR[9]	—	High	—	Low	Low	Post-disaster DTN environment
GeoSaW[10]	—	—	High	High	Low	Airborne DTN search and rescue scenarios
FRLI[11]	—	High	—	Low	—	DTN environment
Optimal flight time algorithm [12]	—	—	High	—	—	DTN environment in smart city scene
Improved TTL-Loc [13]	—	—	High	—	—	DTN environment across multiple islands
DRHT model [14]	High	—	—	Low	Low	DTN environment
MRA[15]	—	—	High	High	Low	Battlefield DTN environment
UAV deployment algorithm [18]	High	—	—	Low	—	Multi-layer UAV relay system
Genetic algorithm [19]	—	—	—	Low	—	Multi-UAV path planning scenario
Rebound flight plan and common flight path plan [23]	—	—	—	Low	—	Battery exhausted UAV flight scene
Weighted minimum and maximum fairness algorithm [26]	—	—	High	Low	—	DTN ferry node storage allocation optimization scenario

V. SUMMARY AND OUTLOOK

In the post-disaster DTN environment, the UAV plays an important role in transmitting messages in the disaster area and communicating effectively with the outside world. The mobility, flexibility and wide coverage of UAVs provide new ideas and methods for data transmission between different partitions of DTN after disaster. However, the UAV network is different from the traditional mobile ad hoc network or the vehicle self-organizing network. The movement in the 3D scene changes rapidly as well as the network topology changes frequently. The energy or storage requirements of the nodes are also need to be considered. This paper combs the relevant literatures on UAV ferry algorithm in DTN environment at home and abroad recently, from the UAV network architecture in DTN, node routing algorithm, node deployment, trajectory optimization, energy optimization and storage allocation, etc. Then this article analyzes the existing methods. There are still many places worthy of research and further optimization for the UAV ferry algorithm in DTN environment. In this paper, combined with the characteristics of the above programs, the following prospects are made:

1) Existing schemes often increase the transmission delay while increasing the message delivery rate and reducing the network cost ratio. A better compromise to effectively perform performance indicators such as message delivery rate, network cost ratio and transmission delay is a further research direction.

2) The existing schemes used for comparative analysis are mostly traditional algorithms and schemes. Introducing the latest research results in the experiment for comparative analysis will promote the application of the new UAV ferry algorithm in DTN more quickly, and will better optimize the performance of related algorithms.

3) While considering the routing algorithm of the UAV ferry nodes, how to comprehensively consider the deployment, trajectory optimization, energy and storage allocation of the UAV nodes is of great importance for synthetically improving the performance of UAV in DTN.

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